Noah Touchton

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EDUCATION

Bachelor of Science in both Mechanical and Aerospace Engineering/Physics Minor Dec 2026
University of Florida, Gainesville, FL GPA: 3.56

Associate of Arts, Mechanical Engineering August 2022 - August 2023
University of Central Florida, Orlando, FL

COURSEWORK

Autonomous Vehicles, Mechanical Design 1, Aerospace Design 1, Controls Lab, Mechanics of Materials Lab, Design and Manufacturing Lab, Compressible Flow, Propulsion, Aerospace Structures, Aerodynamics, Astrodynamics, Differential Equations 2

SKILLS

Proficient in SOLIDWORKS, Arduino, C, C++, Python, ROS2, MATLAB, Xflr5, Microsoft Excel

RESEARCH Oct 2023 - Present

Research Assistant
Smart Autonomy Lab | University of Florida

- Assisted in Dr. Yu Wang’s research on control systems by analyzing the impact of tampered indoor positioning system signals on autonomous vehicle path-following accuracy, using control theory to evaluate performance limits.
- Collected and recorded experimental data to assess the ability of control algorithms to correct distorted location signals, supporting advancements in robust navigation systems.

PROFESSIONAL EXPERIENCE

Mechanical Engineering Intern
Advanced Autonomy Specialists Inc. | Jacksonville, FL May 2025 – Present

- Configured ROS2 on Jetson Nano/JetBot with workspace setup and Gazebo integration.
- Integrated object detection, behavior trees, state machines, and action servers so the JetBot can find and follow RC Cars.
- Developed and simulated a ROS2 Context Understanding Engine using ontology-driven reasoning and behavior-tree deliberation.

INVOLVEMENT

Motion Simulator Design Team August 2024 – Present
Theme Park Engineering and Design Club | University of Florida

- Working on the software team to design and develop a user interface for controlling an Arduino-driven motion chair with three-axis pivoting.
- Applying control theory to synchronize software inputs with the real-time motion of the chair.
- Writing and integrating Arduino and Python code to manage the motion control system.

Aerodynamics Sub-team August 2024 – Present
Design Build Fly Competition | University of Florida

- Contributing to the design and optimization of aircraft aerodynamics for the Design Build Fly competition.
- Utilizing XFLR5 and other aerodynamic analysis tools to refine wing design, performance, and stability.

Animatronic Design Team January 2024 – Present
Theme Park Engineering and Design Club | University of Florida

- Developed software to control the motion of an animatronic gator head, synchronizing the chomping mechanism with sensor inputs.
- Programmed Arduino microcontrollers to manage the gator head's movements and respond to external triggers.
- Collaborated with hardware and mechanical teams to integrate the software with the animatronic system, ensuring smooth operation.